

Problem-Solution Fit

Date	29 June 26
Team ID	xxxxxx
Project Name	Opticrop: Smart Agricultural production optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Small and medium farmers, agricultural cooperatives, and agri-businesses seeking data-driven crop optimization.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited internet access, low digital literacy, budget constraints, and lack of smart devices in rural areas.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional farming methods, weather apps, and manual crop advisory services. Pros: Easy to access. Cons: Not personalized and less accurate.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Unpredictable crop yield, improper irrigation, pest attacks, and inefficient resource utilization occurring every season.	9. PROBLEM ROOT / CAUSE [RC] Lack of real-time agricultural insights, climate variability, and inadequate crop monitoring systems.	7. BEHAVIOR + ITS INTENSITY [BE] Farmers frequently rely on experience-based decisions and require timely recommendations during each crop cycle.
3. TRIGGERS TO ACT [TR] Declining productivity, rising input costs, water scarcity, and market demand fluctuations motivate farmers to adopt technology.	10. YOUR SOLUTION [SL] Opticrop uses AI, sensor data, and analytics to optimize irrigation, fertilizer usage, and crop management recommendations. Provides predictive analytics, disease detection, and smart recommendations for sustainable agriculture.	8. CHANNELS OF BEHAVIOR [CH] ONLINE ONLINE: Mobile applications, agricultural websites, SMS alerts, and social media farming communities. OFFLINE OFFLINE: Agricultural extension centers, farmer meetings, training camps, and local cooperatives.
4. EMOTIONS BEFORE / AFTER [EM] Before: Stress, uncertainty, and financial risk. After: Confidence, improved		

productivity, and better decision-making.

